Problem 1: A magic square is a square matrix where the sum of each row, each column, and both diagonals are equal. Let W be the set of all 3×3 real-valued magic squares.

Is W a subspace of the vector space of all 3×3 real matrices? Justify your answer.

Solution

To determine if W, the set of 3×3 real-valued magic squares, is a subspace of $M_{3\times3}(\mathbb{R})$, we verify the subspace criteria.

1. **Zero Matrix**: The zero matrix $O = [0]_{3\times3}$ has all row, column, and diagonal sums equal to 0, so $O \in W$.

2. **Addition**: Let $A, B \in W$ with magic sum s. For $A + B = [a_{ij} + b_{ij}]$:

   - Row sum: $\sum_{j=1}^{3}(a_{ij} + b_{ij}) = \sum_{j=1}^{3} a_{ij} + \sum_{j=1}^{3} b_{ij} = s + s = 2s$.
   - Column sum: $\sum_{i=1}^{3}(a_{ij} + b_{ij}) = s + s = 2s$.
   - Diagonal sums: $\sum_{i=1}^{3}(a_{ii} + b_{ii}) = s + s = 2s$, and similarly for the other diagonal.

   Thus, $A + B \in W$.

3. **Scalar Multiplication**: Let $A \in W$ with magic sum s and $k \in \mathbb{R}$. For $kA = [ka_{ij}]$:

   - Row sum: $\sum_{j=1}^{3} ka_{ij} = k\sum_{j=1}^{3} a_{ij} = ks$.
   - Column sum: $\sum_{i=1}^{3} ka_{ij} = ks$.
   - Diagonal sums: $\sum_{i=1}^{3} ka_{ii} = ks$, and similarly for the other diagonal.

   Thus, $kA \in W$.

Since W contains the zero matrix and is closed under addition and scalar multiplication, W is a subspace of $M_{3\times3}(\mathbb{R})$.

Problem 2: Find $\mathbf{u} \cdot \mathbf{v}$ given that $\|\mathbf{u} + \mathbf{v}\| = 3$ and $\|\mathbf{u} - \mathbf{v}\| = 1$.

Solution

Recall that for any vector $\mathbf{w}$, the norm satisfies $\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w}$. Thus:

(i) $\|\mathbf{u} + \mathbf{v}\|^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = \|\mathbf{u}\|^2 + 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2 = 9,$

(ii) $\|\mathbf{u} - \mathbf{v}\|^2 = (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = \|\mathbf{u}\|^2 - 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2 = 1.$

Subtract (ii) from (i):

$$(\|\mathbf{u}\|^2 + 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2) - (\|\mathbf{u}\|^2 - 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2) = 9 - 1$$

$$4\mathbf{u} \cdot \mathbf{v} = 8$$

$$\mathbf{u} \cdot \mathbf{v} = 2.$$

Conclusion: The dot product is:

$$\boxed{2}$$

Problem 3: Find two vectors of norm 1 that are orthogonal to $\mathbf{u} = (2, 1, -4, 0)$ and $\mathbf{v} = (-1, -1, 2, 2)$.

Solution

Consider $\mathbf{w} = (w_1, w_2, w_3, w_4)$ such that:

(i) $\mathbf{u} \cdot \mathbf{w} = 2w_1 + w_2 - 4w_3 = 0$,

(ii) $\mathbf{v} \cdot \mathbf{w} = -w_1 - w_2 + 2w_3 + 2w_4 = 0$,

(iii) $\|\mathbf{w}\| = 1$.

From (i): $w_2 = -2w_1 + 4w_3$. Substitute into (ii):

$$-w_1 - (-2w_1 + 4w_3) + 2w_3 + 2w_4 = w_1 - 2w_3 + 2w_4 = 0 \implies w_1 = 2w_3 - 2w_4,$$

$$w_2 = -2(2w_3 - 2w_4) + 4w_3 = 4w_4.$$

Thus, $\mathbf{w} = (2w_3 - 2w_4, 4w_4, w_3, w_4)$. Set $s = w_3$, $t = w_4$:

$$\mathbf{w} = (2s - 2t, 4t, s, t).$$

Norm: $\|\mathbf{w}\|^2 = 5s^2 - 8st + 21t^2 = 1$.
Case 1: $t = 0$, $s = \frac{1}{\sqrt{5}}$:

$$\mathbf{w}_1 = \left(\frac{2}{\sqrt{5}}, 0, \frac{1}{\sqrt{5}}, 0\right), \quad \|\mathbf{w}_1\|^2 = \frac{4}{5} + \frac{1}{5} = 1.$$

Case 2: $s = 0$, $t = \frac{1}{\sqrt{21}}$:

$$\mathbf{w}_2 = \left(-\frac{2}{\sqrt{21}}, \frac{4}{\sqrt{21}}, 0, \frac{1}{\sqrt{21}}\right), \quad \|\mathbf{w}_2\|^2 = \frac{4}{21} + \frac{16}{21} + \frac{1}{21} = 1.$$

Verify: $\mathbf{u} \cdot \mathbf{w}_1 = 0$, $\mathbf{v} \cdot \mathbf{w}_1 = 0$, $\mathbf{u} \cdot \mathbf{w}_2 = 0$, $\mathbf{v} \cdot \mathbf{w}_2 = 0$.

$$\boxed{\left(\frac{2}{\sqrt{5}}, 0, \frac{1}{\sqrt{5}}, 0\right), \left(-\frac{2}{\sqrt{21}}, \frac{4}{\sqrt{21}}, 0, \frac{1}{\sqrt{21}}\right)}$$

Problem 4: Let V be the vector space of all 3×3 real matrices. Define

$$W = \{A \in V \mid \det(A) = 0\}.$$

Is W a subspace of V? Justify your answer.

Solution

Check the subspace criteria:
1. **Zero Element**: The zero matrix $O \in V$ has $\det(O) = 0$, so $O \in W$.
2. **Closure under Addition**: Consider $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, both in W since $\det(A) = 0 \cdot 1 \cdot 1 = 0$ and $\det(B) = 0 \cdot 0 \cdot 1 = 0$. Compute:

$$A + B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \det(A + B) = 1 \cdot 1 \cdot 1 = 1 \neq 0.$$

Thus, $A + B \notin W$, so W is not closed under addition.
3. **Closure under Scalar Multiplication**: For $A \in W$ and $c \in \mathbb{R}$, $\det(cA) = c^3 \det(A) = c^3 \cdot 0 = 0$, so $cA \in W$.
Since W is not closed under addition, it is not a subspace.

$\boxed{W \text{ is not a subspace of } V}$

Problem 5: Let V be the vector space of all 3×3 real matrices. Define

$$W = \{A \in V \mid A^T = A\}.$$

Is W a subspace of V? Justify your answer.

Solution

Verify the subspace criteria:
1. **Zero Element**: The zero matrix O satisfies $O^T = O$, so $O \in W$.
2. **Closure under Addition**: Let $A, B \in W$, so $A^T = A$ and $B^T = B$. Then:

$$(A + B)^T = A^T + B^T = A + B,$$

so $A + B \in W$.
3. **Closure under Scalar Multiplication**: For $A \in W$ and $k \in \mathbb{R}$:

$$(kA)^T = kA^T = kA,$$

so $kA \in W$.
Since all criteria hold, W is a subspace.

$\boxed{W \text{ is a subspace of } V}$

Problem 6: Let V be the vector space of all 3×3 real matrices. Define

$$W = \{A \in V \mid A^T = -A\}.$$

Is W a subspace of V? Justify your answer.

Solution

Verify the subspace criteria:
1. **Zero Element**: The zero matrix O satisfies $O^T = O = -O$, so $O \in W$.
2. **Closure under Addition**: Let $A, B \in W$, so $A^T = -A$ and $B^T = -B$. Then:

$$(A + B)^T = A^T + B^T = -A + (-B) = -(A + B),$$

so $A + B \in W$.
3. **Closure under Scalar Multiplication**: For $A \in W$ and $k \in \mathbb{R}$:

$$(kA)^T = kA^T = k(-A) = -kA,$$

so $kA \in W$.
Since all criteria hold, W is a subspace.

$\boxed{W \text{ is a subspace of } V}$

Problem 7: Let $V = C([a, b])$ be the vector space of all continuous functions on the closed interval $[a, b]$. Define the subset

$$W = \left\{ f \in V \mid \int_a^b f(x)\, dx = 0 \right\}.$$

Is W a subspace of V. Justify your answer.

Solution

Verify the subspace criteria:

1. **Zero Element**: The zero function $f(x) = 0$ satisfies $\int_a^b 0\, dx = 0$, so $0 \in W$.
2. **Closure under Addition**: Let $f, g \in W$, so $\int_a^b f(x)\, dx = 0$ and $\int_a^b g(x)\, dx = 0$. Then:

$$\int_a^b (f(x) + g(x))\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx = 0 + 0 = 0,$$

so $f + g \in W$.

3. **Closure under Scalar Multiplication**: For $f \in W$ and $k \in \mathbb{R}$:

$$\int_a^b (kf(x))\, dx = k \int_a^b f(x)\, dx = k \cdot 0 = 0,$$

so $kf \in W$.

Since all criteria hold, W is a subspace.

$$\boxed{W \text{ is a subspace of } V}$$